Description about JHU Multi-Atlas library

1: Overview:

For quantitative analysis of brain MR images, one of the first and most critical steps is to define corresponding structures across subjects. One of the most widely used approaches is the voxel-based approach, in which shapes and positions of all subjects are aligned such that anatomical locations can be references by x, y, z coordinates in a standardized space such as MNI coordinates. An alternative way is based on structural segmentation. For example, we can define the hippocampus by grouping all voxels that belong to the hippocampus in each subject. Then we can compare the voxel-aggregated information, such as the number of voxels (volume) and average voxel intensity, across different subjects. The latter approach, however, requires the human knowledge about the definition of the hippocampus. There are many publicly-available automated segmentation tools and, to our best knowledge, all of them are based on internal or external brain atlases that were based on, at some point, manual structural delineation by neuroanatomists.

In recent years, a new automated image segmentation approaches, called Multi-Atlas Label Fusion, have advanced considerably. This approach, however, requires a large number of pre-segmented atlases which are not readily available. In this atlas resource, we provide fully segmented brain atlases that cover 4 – 90 years old subjects and contain 289 structural labels (Version 7.1). This document describes the details of the structures defined in these atlases.

2: Creation of initial single-brain atlas:

Anatomical definition cannot always be done with clear biological criteria because anatomy does not always have clear boundaries. For example, for the definition of the hippocampus, a recent survey about protocols for the manual delineation found as much as 250% of volume differences (Boccardi et al., 2011). For segmentation-based analysis, therefore, it is essential to establish consistent anatomical criteria to demarcate structures of interest. For this purpose, a single-brain atlas with manually defined structure often serves as a reference. In this multi-atlas library, we used the JHU-MNI single-subject atlas (also known as “Eve” atlas) as our anatomical reference, which was initially described by Oishi et al (Oishi et al., 2009).

<u>2-1: MRI data:</u> The single-subject atlas consisted of co-registered T1, T2, and DTI data acquired using a 1.5 T MR scanner (Gyroscan NT, Philips Medical Systems). Data were obtained from a 32-year-old healthy female participant without known brain or neurological disorders. The T1-weighted image was based on a MPRAGE sequence with 1 mm isotropic resolution. T2-weighted image was acquired using a double-echo (40 / 100 ms) Fast Spin-Echo (FSE) sequence with 1x1x2 mm resolution. DTI data were acquired using single-shot echo-planar imaging sequences with sensitivity encoding (SENSE) and a parallel imaging factor of 2.5 (Pruessmann et al., 1999). The imaging matrix was 112×112 (60 slices), with a field of view of 246×246 mm (nominal resolution: 2.2 mm), zero-filled to 256×256 pixels. Transverse sections were acquired parallel to the anterior commissure–posterior commissure line (AC–PC). Diffusion weighting was encoded along 30 independent orientations (Jones et al., 1999), and the b-value was 700 mm2/s. Five additional images with minimal diffusion weighting (b=33 mm2/s) were also acquired. All images were co-registered in the MNI space with 1x1x1 mm resolution and a 181x217x181 matrix size.

<u>2-2: Gray matter parcellation:</u> The segmentation of the gray matter structures were performed by Drs. Kenichi Oishi and Susumu Mori. The definition of the gray matter followed LONI Probabilistic Brain Atlas (LPBA40) (Shattuck et al., 2008). The probabilistic locations of 62 gray matter structures, including 52 cortical parcels, defined in atlas were transferred to the single-subject atlas and the detailed boundaries were defined manually in the JHU-MNI single-subject atlas. The detailed description of the gray matter is provided in Appendix I.

<u>2-3: Deep white matter segmentation:</u> All white matter segmentation was performed by manual delineation by Dr. Susumu Mori. The criteria for the core white matter segmentation followed the probabilistic white matter delineation described by (Mori et al., 2008), in which 56 deep white matter areas were defined. The detailed description of the deep white matter is provided in Appendix II.

2-4: Superficial white matter segmentation: The 46 superficial white matter segments, that locate between the deep white matter and the cortical regions were defined based on the probabilistic white matter atlas described in (Oishi et al., 2008). The detailed description of the superficial white matter is provided in Appendix I.

2-5: Other structural labels: The atlas also contains 12 other structures including ventricles and structures in the brainstem and the cerebellum. The total number of the structural labels was 176.

3: Additions and modifications of structural labels

The structural labeling underwent several modifications. The version released to the public as of 5/1/15 was Version 7.1, which contains 289 structures. The complete list of defined structures is provided in Appendix III. Also, all abbreviations are defined in Appendix IV. The added structures include finer divisions of the ventricle system and deep gray matter nuclei. The anatomical criteria of the additional deep gray matter structures followed the labeling in "Atlas of Human Brain" by Mai & Paxinos. (Mai et al., 2007)

4: Generation of initial 29 adult atlases

The manual labeling of the structures takes prohibitively long and it is not practical to perform for a large number of images. For the subsequent data, an atlas-based automated segmentation was used. In this approach, the single-subject atlas with the structural labels was used as an anatomical reference and warped to 29 adult brain images using a method described in (Djamanakova et al., 2013). To drive the warping, MPRAGE images were used. Elastic warping by Large Deformation Diffeomorphic Metric Mapping (LDDMM) and two channels (images and manually segmented ventricles) were used. The resultant transformation matrix was applied to the co-registered structure labeling files, which automatically segment the target images. Errors in the segmentation results were then manually corrected by an expert (Dr. Susumu Mori), following the same criteria applied to the first single-subject atlas.

5: Generation of the large atlas library

In addition to MRI data from JHU (8-85 years old), the following publicly available data were used to expand the atlas library.

ADNI (http://adni.loni.usc.edu/) / 66-92 years old
Kirby21 (https://www.nitrc.org/projects/multimodal/) / 23-62 years old
PING (http://pingstudy.ucsd.edu/) / 4-20 years old

Using the initial 29 multi-atlas library and multi-atlas segmentation algorithms (Tang et al., 2014), an adult library (> 20 years old) was first created and all segmentation results were manually corrected to conform to the anatomical definitions described above. To establish multi-atlas libraries for younger populations, existing atlases with a similar age range were used. Namely, using the young adult data (20-42 years old), pediatric data (8-20 years old) were segmented and manually corrected. The 4-7 years old data were then segmented using 8-12 years old data and manually corrected. The manual corrections were performed by Dr. Susumu Mori, but for consistently observed errors, which often happens for cross-age automated segmentation, pictorial protocols were established and multiple members in Dr. Mori participated for the correction. Following protocols for corrections of common errors and addition of new labels for newer versions are available at www.mricloud.org.

5-1: Correction protocols for cortical definitions:

- CorticalLeakCorrection.ppt
- CommonDuraError.ppt

5-2: Correction protocols for pediatric atlas libraries;

- PedCortexFix.ppt
- CommonPedErrorCorrection.ppt
- PedCerebellumCorrection.ppt
- PedCaudateCorrection.ppt

5-3: Protocols for new versions

- AtlasV6a.ppt
- AtlasV6b.ppt
- AtlasV7a.ppt

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Appendix I: Parcellation of the gray matter and peripheral (cortical) white matter

We used maximum likelihood maps, which identify the most likely gyrus at each voxel in the atlas space, provided by the Laboratory of Neuro Imaging (LONI), University of California, Los Angeles. This is called The LONI Probabilistic Brain Atlas (LPBA40) and can be downloaded from the website (http://www.loni.ucla.edu/Atlases/). The atlas contains 24 gyri labels. It also defines 6 deep gray matter structures, the brainstem, and the cerebellum. For the hippocampus, the detailed delineation criteria followed a paper and actual images by Haller et al (Haller et al., 1997). For the Amygdala, the criteria currently follows the description by Entis et al (Entis et al., 2012).

For the 21 gyri in Table 1, the white matter areas between the cortex and the deep white matter structures were defined as peripheral white matter areas. The insula, parahippocampal, and cingulate gyri face directly to the deep white matter structures and, thus, no peripheral white matter structures associated these gyri were defined.

Table 1: The cortical parcellation defined in this table has both gray matter (cortex) and white matter (peripheral white matter) labels

Lobes	Subparcellation
Frontal lobe	Superior frontal, rectus, middle orbito-frontal
Frontal lobe	Middle frontal
Frontal lobe	Inferior frontal, lateral orbito-frontal
Frontal lobe	Pre-central
Parietal lobe	Post-central
Parietal lobe	Superior parietal, precuneus
Parietal and temporal lobe	Supramarginal, angular, superior temporal W
Temporal lobe	Middle temporal, inferior temporal, anterior part of fusiform
Occipital lobe	Superior occipital, middle occipital, inferior occipital, cuneus, lingual, posterior part of fusiform

Appendix II: Parcellation of the deep white matter

(1) Tracts in the brainstem:

Corticospinal tracts (CST): This structure can be clearly identified at the medulla and the pons level, but should also contain corticopontine and corticobulbar tracts

Medial lemniscus (ML): This tract is identifiable in the pons, but not in the midbrain. Because of the limited image resolution, this parcellation may include the central tegmental tract.

Inferior cerebellar peduncle (ICP): This tract can be identified from the medulla level toward the cerebellum.

Middle cerebellar peduncle (MCP): This massive tract initiates from the pontine nuclei and carries information between the cortex and the cerebellum. In the pons, this segment orienting along the right-left direction was specifically segmented as the pontine crossing tract (PCT). Also a portion of MCP inside the cerebellum was labels as “MCP in the cerebellar section”.

Superior cerebellar peduncle (SCP): This tract is identifiable from the cerebellar nuclei to the midbrain at the SCP deccusation. After the deccusation, the tract cannot be identified with current image resolution.

(2) Projection fibers

Corona radiata: This structure is divided into three regions: anterior (ACR), superior (SCR), and posterior (PCR). The divisions are made at the middle of the genu and splenium of the corpus callosum, which are arbitrarily chosen and not based on anatomic or functional boundaries. The boundary of the corona radiata and the internal capsule is defined at the axial level where the internal capsule and the external capsule merge.

Anterior limb of internal capsule (ALIC): This white matter is defined as a structure between the caudate and the putamen.

Posterior limb of internal capsule (PLIC): This white matter is defined as a structure between the Thalamus and the globus pallidus.

Retrolenticular part of the internal capsule (RLIC): In this region, the posterior thalamic radiation (corticothalamic and thalamocortical fibers, including the optic radiation) is the major constituent, but can also include the parieto-, occipito- and temporopontine fibers. The boundary with the sagittal stratum (SS) is arbitrarily defined at the middle of the splenium of the corpus callosum.

Cerebral peduncle (CP): This is a region where long corticofugal pathways are concentrated, including the corticospinal, corticopontine, and corticobulbar tracts. The boundary between the cerebral peduncle and the internal capsule is defined at the axial level below the anterior commissure

(3) Association fibers

Superior longitudinal fasciculus (SLF): This tract is located at the dorsolateral regions of the corona radiata and contains connections between the frontal, parietal, occipital, and temporal lobes including language-related areas (Broca's, Geschwind's, and Wernicke's territories).

Superior fronto-occipital fasciculus (SFO): This tract is located at the superior edge of the anterior limb of the internal capsule (anterior thalamic radiation) and the boundary is not always clear. Only the frontal region is identifiable and projection to the parietal lobe cannot be segmented.

Uncinate fasciculus (UNC): This tract connects the frontal lobe (orbital cortex) and the anterior temporal lobe. It can be discretely identified where the two lobes are connected but not within the frontal and the temporal lobes where it merges with other tracts.

Inferior fronto-occipital fasciculus (IFO): The IFO connects the frontal lobe and the occipital lobe. In the frontal lobe, this partition also includes the frontal projection of the UNC. In the temporal and occipital lobe, the IFO merges with the inferior longitudinal fasciculus (ILF), which is segmented as a different partition.

Inferior longitudinal fasciculus (ILF): This partition includes the white matter in the temporal and occipital lobe where the IFO and the ILF are the major constituents. The ILF connects the temporal lobe and the occipital lobes. It cannot be distinguished from the IFO in most of the temporal and occipital white matter.

Sagittal stratum (SS): The IFO/ILF merges with projection fibers from the RLIC and forms a large, sheet-like, sagittal structure, called the sagittal stratum. This region, therefore, should include both association and projection fibers. The boundary of the IFO/ILF and SS is arbitrarily defined at the axial level of the anterior commissure. The boundary of the SS and the PCRis also arbitrarily defined at the axial level of the splenium of the corpus callosum.

External capsule (EC): This region, located lateral to the internal capsule and adjacent to the insula is believed to contain association fibers, such as the SLF and IFO and commissural fibers. Because of the limited image resolution, the external and extreme capsules are not resolved.

Cingulum (CG): This tract carries information from the cingulate gyrus to the hippocampus. The entire pathway from the frontal lobe to the temporal lobe can be clearly identified. In the WMPM, the CG in the cingulate gyrus and the hippocampal regions is separated at the axial level of the splenium of the corpus callosum and denoted as CgC and CgH, respectively.

Fornix (FX) and stria terminalis (ST): These tracts are both related to the limbic system: the FX to the hippocampus, and the ST to the amygdala. Both tracts project to the septum and the hypothalamus. With current image resolution capabilities, these two tracts cannot be distinguished in the hippocampal area, and both tracts are labeled as FX. The ST can be discretely identified in the amygdala and the dorsal thalamus.

(4) Commissural fibers

Anterior commissure (AC): The projection to the temporal lobes of the AC is segmented. Corpus callosum (CC): This partition contains the corpus callosum and the boundary extends until it merges with the corona radiata. The CC is further divided into the genu (GCC), the body (BCC), and the splenium (SCC) regions with arbitrary boundaries.
Tapetum (TAP): This temporal component of the CC is partitioned separately from the other CC regions.

Appendix III: Structures defined in Version 7.1

1 SFG_L
2 SFG_R
3 SFG_PFC_L
4 SFG_PFC_R
5 SFG_pole_L
6 SFG_pole_R
7 MFG_L
8 MFG_R
9 MFG_DPFC_L
10 MFG_DPFC_R
11 IFG_opercularis_L
12 IFG_opercularis_R
13 IFG_orbitalis_L
14 IFG_orbitalis_R
15 IFG_triangularis_L
16 IFG_triangularis_R
17 LFOG_L
18 LFOG_R
19 MFOG_L
20 MFOG_R
21 RG_L
22 RG_R
23 PoCG_L
24 PoCG_R
25 PrCG_L
26 PrCG_R
27 SPG_L
28 SPG_R
29 SMG_L
30 SMG_R
31 AG_L
32 AG_R
33 PrCu_L
34 PrCu_R
35 STG_L
36 STG_R
37 STG_L_pole
38 STG_R_pole
39 MTG_L
40 MTG_R
41 MTG_L_pole
42 MTG_R_pole
43 ITG_L
44 ITG_R
45 PHG_L
46 PHG_R
47 ENT_L

48 ENT_R
49 FuG_L
50 FuG_R
51 SOG_L
52 SOG_R
53 MOG_L
54 MOG_R
55 IOG_L
56 IOG_R
57 Cu_L
58 Cu_R
59 LG_L
60 LG_R
61 rostral_ACC_L
62 rostral_ACC_R
63 subcallosal_ACC_L
64 subcallosal_ACC_R
65 subgenual_ACC_L
66 subgenual_ACC_R
67 dorsal_ACC_L
68 dorsal_ACC_R
69 PCC_L
70 PCC_R
71 Insula_L
72 Insula_R
73 Amyg_L
74 Amyg_R
75 Hippo_L
76 Hippo_R
77 Caud_L
78 Caud_R
79 Put_L
80 Put_R
81 GP_L
82 GP_R
83 Thalamus_L
84 Thalamus_R
85 HypoThalamus_L
86 HypoThalamus_R
87 BasalForebrain_L
88 BasalForebrain_R
89 NucAccumbens_L
90 NucAccumbens_R
91 RedNc_L
92 RedNc_R
93 Snigra_L
94 Snigra_R
95 CerebellumGM_R

96 CerebellumGM_L
97 CP_L
98 CP_R
99 Midbrain_L
100 Midbrain_R
101 CST_L
102 CST_R
103 SCP_L
104 SCP_R
105 MCP_L
106 MCP_R
107 PCT_L
108 PCT_R
109 ICP_L
110 ICP_R
111 ML_L
112 ML_R
113 Pons_L
114 Pons_R
115 Medulla_L
116 Medulla_R
117 ACR_L
118 ACR_R
119 SCR_L
120 SCR_R
121 PCR_L
122 PCR_R
123 GCC_L
124 GCC_R
125 BCC_L
126 BCC_R
127 SCC_L
128 SCC_R
129 PVWl_L
130 PVWl_R
131 ALIC_L
132 ALIC_R
133 PLIC_L
134 PLIC_R
135 RLIC_L
136 RLIC_R
137 EC_L
138 EC_R
139 CGC_L
140 CGC_R
141 CGH_L
142 CGH_R
143 Fx/ST_L

144 Fx/ST_R
145 Fx_L
146 Fx_R
147 IFO_L
148 IFO_R
149 PTR_L
150 PTR_R
151 SS_L
152 SS_R
153 SFO_L
154 SFO_R
155 SLF_L
156 SLF_R
157 Cl_L
158 Cl_R
159 BasalForebrain_L
160 BasalForebrain_R
161 BasalForebrain_L
162 BasalForebrain_R
163 BasalForebrain_L
164 BasalForebrain_R
165 BasalForebrain_L
166 BasalForebrain_R
167 Mammillary_R
168 Mammillary_L
169 OpticTract_L
170 OpticTract_R
171 LV_Frontal_L
172 LV_body_L
173 LV_atrium_L
174 LV_Occipital_L
175 LV_Inferior_L
176 LV_Frontal_R
177 LV_body_R
178 LV_atrium_R
179 LV_Occipital_R
180 LV_Inferior_R
181 III_ventricle
182 PVWa_L
183 PVWa_R
184 PVWp_L
185 PVWp_R
186 SFWM_L
187 SFWM_R
188 SFWM_PFC_L
189 SFWM_PFC_R
190 SFWM_pole_L
191 SFWM_pole_R

192 MFWM_L
193 MFWM_R
194 MFWM_DPFC_L
195 MFWM_DPFC_R
196 IFWM_opercularis_L
197 IFWM_opercularis_R
198 IFWM_orbitalis_L
199 IFWM_orbitalis_R
200 IFWM_triangularis_L
201 IFWM_triangularis_R
202 LFOWM_L
203 LFOWM_R
204 MFOWM_L
205 MFOWM_R
206 RGWM_L
207 RGWM_R
208 PoCWM_L
209 PoCWM_R
210 PrCWM_L
211 PrCWM_R
212 SPWM_L
213 SPWM_R
214 SMWM_L
215 SMWM_R
216 AGWM_L
217 AGWM_R
218 PrCuWM_L
219 PrCuWM_R
220 STWM_L
221 STWM_R
222 STWM_L_pole
223 STWM_R_pole
224 MTWM_L
225 MTWM_R
226 MTWM_L_pole
227 MTWM_R_pole
228 ITWM_L
229 ITWM_R
230 FuWM_L
231 FuWM_R
232 SOWM_L
233 SOWM_R
234 MOWM_L
235 MOWM_R
236 IOWM_L
237 IOWM_R
238 CuWM_L
239 CuWM_R

240 LWM_L
241 LWM_R
242 rostralWM_ACC_L
243 rostralWM_ACC_R
244 subcallosalWM_ACC_L
245 subcallosalWM_ACC_R
246 subgenualWM_ACC_L
247 subgenualWM_ACC_R
248 dorsalWM_ACC_L
249 dorsalWM_ACC_R
250 PCCWM_L
251 PCCWM_R
252 CerebellumWM_R
253 CerebellumWM_L
254 CSF
255 SKULL1
256 SKULL2
257 SKULL3
258 MCP_cb_L
259 MCP_cb_R
260 Bone marrow
261 ICP_pons_L
262 ICP_pons_R
263 FrontSul_L
264 FrontSul_R
265 CentralSul_L
266 CentralSul_R
267 SylFrontSul_L
268 SylFrontSul_R
269 SylTempSul_L
270 SylTempSul_R
271 SylParieSul_L
272 SylParieSul_R
273 ParietSul_L
274 ParietSul_R
275 CinguSul_L
276 CinguSul_R
277 OcciptSul_L
278 OcciptSul_R
279 TempSul_L
280 TempSul_R
281 Caudate_tail_L
282 Fimbria_L
283 Caudate_tail_R
284 Fimbria_R
285 Chroid_LVetc_L
286 Chroid_LVetc_R
287 IV_ventricle

ECCL_L
ECCL_R

Appendix IV: Abbreviations

ACR_L	Anterior corona radiata left
ACR_R	Anterior corona radiata right
AG_L	ANGULAR GYRUS left
AG_R	ANGULAR GYRUS right
ALIC_L	Anterior limb of internal capsule left
ALIC_R	Anterior limb of internal capsule right
Amyg_L	AMYGDALA left
Amyg_R	AMYGDALA right
AnsaLenticularis_L	Ansa lenticularis left
AnsaLenticularis_R	Ansa lenticularis right
AnteriorCom_L	Anterior commissure left
AnteriorCom_R	Anterior commissure right
AWM_L	ANGULAR WM left
AWM_R	ANGULAR WM right
BCC_L	Body of corpus callosum left
BCC_R	Body of corpus callosum right
Bone marrow	
Caud_L	CAUDATE NUCLEUS left
Caud_R	CAUDATE NUCLEUS right
CaudT_L	Caudate Tail Left
CaudT_R	Caudate Trail Right
CentralSul_L	Central Sulcus Left
CentralSul_R	Central Sulcus Right
CerebellumGM_L	CEREBELLUM left
CerebellumGM_R	CEREBELLUM right
CerebellumWM_L	cerebellum WM left
CerebellumWM_R	cerebellum WM right
CGC_L	Cingulum (cingulate gyrus) left
CGC_R	Cingulum (cingulate gyrus) right
CGH_L	Cingulum (hippocampus) left
CGH_R	Cingulum (hippocampus) right
CinguSul_L	Cingular Cortex Sulci Left
CinguSul_R	Cingular Cortex Sulci Right
Cl_L	Clustrum Complex Left
Cl_R	Clustrum Complex RIght
CP_L	Cerebral peduncle left
CP_R	Cerebral peduncle right
CSF	Peripheral CSF
CST_L	Corticospinal tract left
CST_R	Corticospinal tract right
Cu_L	CUNEUS left
Cu_R	CUNEUS right
CuWM_L	CUNEUS WM left
CuWM_R	CUNEUS WM right
dorsal_ACC_L	dorsal anterior cingulate gyrus left
dorsal_ACC_R	dorsal anterior cingulate gyrus right
dorsalWM_ACC_L	dorsal anterior cingulate WM left

dorsalWM_ACC_R	dorsal anterior cingulate WM right
EC_L	External capsule left
EC_R	External capsule right
ECCL_L	External capsule / claustrum left
ECCL_R	External capsule / Claustrum right
ENT_L	ENTORHINAL AREA left
ENT_R	ENTORHINAL AREA right
Fimbria_L	Fimbria left
Fimbria_R	Fimbria right
FrontSul_L	Frontal Lobe Sulci Left
FrontSul_R	Frontal Lobe Sulci Right
FuG_L	FUSIFORM GYRUS left
FuG_R	FUSIFORM GYRUS right
FuWM_L	FUSIFORM WM left
FuWM_R	FUSIFORM WM right
Fx/ST_L	Fornix (cres) / Stria terminalis (can not be resolved with current resolution) left
Fx/ST_R	Fornix (cres) / Stria terminalis (can not be resolved with current resolution) right
Fx_L	Fornix (column and body of fornix) left
Fx_R	Fornix (column and body of fornix) right
GCC_L	Genu of corpus callosum left
GCC_R	Genu of corpus callosum right
GP_L	GLOBUS PALLIDUS left
GP_R	GLOBUS PALLIDUS right
Hippo_L	HIPPOCAMPUS left
Hippo_R	HIPPOCAMPUS right
HypoThalamus_L	Hypothalamus left
HypoThalamus_R	Hypothalamus right
ICP_L	Inferior cerebellar peduncle left
ICP_pons_L	Inferior Cerebellar Peduncle Pons Part left
ICP_pons_R	Inferior Cerebellar Peduncle Pons Part right
ICP_R	Inferior cerebellar peduncle right
IFG_opercularis_L	inferior frontal gyrus pars opercularis left
IFG_opercularis_R	inferior frontal gyrus pars opercularis right
IFG_orbitalis_L	inferior frontal gyrus pars orbitralis left
IFG_orbitalis_R	inferior frontal gyrus pars orbitralis right
IFG_triangularis_L	inferior frontal gyrus pars triangularis left
IFG_triangularis_R	inferior frontal gyrus pars triangularis right
IFO_L	Inferior fronto-occipital fasciculus left
IFO_R	Inferior fronto-occipital fasciculus right
IFWM_opercularis_L	inferior frontal WM pars opercularis left
IFWM_opercularis_R	inferior frontal WM pars opercularis right
IFWM_orbitalis_L	inferior frontal WM pars orbitralis left
IFWM_orbitalis_R	inferior frontal WM pars orbitralis right
IFWM_triangularis_L	inferior frontal WM pars triangularis left
IFWM_triangularis_R	inferior frontal WM pars triangularis right
III_ventricle	Third ventricle
IV_ventricle	Fourth ventricle
Ins_L	INSULAR left

Ins_R	INSULAR right
IOG_L	INFERIOR OCCIPITAL GYRUS left
IOG_R	INFERIOR OCCIPITAL GYRUS right
IOWM_L	INFERIOR OCCIPITAL WM left
IOWM_R	INFERIOR OCCIPITAL WM right
ITG_L	INFERIOR TEMPORAL GYRUS left
ITG_R	INFERIOR TEMPORAL GYRUS right
ITWM_L	INFERIOR TEMPORAL WM left
ITWM_R	INFERIOR TEMPORAL WM right
LenticularFasc_L	Lenticular fasciclus left
LenticularFasc_R	Lenticular fasciclus right
LFOG_L	LATERAL FRONTO-ORBITAL GYRUS left
LFOG_R	LATERAL FRONTO-ORBITAL GYRUS right
LFOWM_L	LATERAL FRONTO-ORBITAL WM left
LFOWM_R	LATERAL FRONTO-ORBITAL WM right
LG_L	LINGUAL GYRUS left
LG_R	LINGUAL GYRUS right
LV_atrium_L	lateral ventricle_atrium left
LV_atrium_R	lateral ventricle_atrium right
LV_body_L	lateral ventricle_body left
LV_body_R	lateral ventricle_body right
LV_Frontal_L	lateral ventricle_frontal left
LV_Frontal_R	lateral ventricle_frontal right
LV_Occipital_L	lateral ventricle_occipital left
LV_Occipital_R	lateral ventricle_occipital right
LV_Temporal_L	lateral ventricle_temporal left
LV_Temporal_R	lateral ventricle_temporal right
LWM_L	LINGUAL WM left
LWM_R	LINGUAL WM right
Mammillary_L	Mammillary body right
Mammillary_R	Mammillary body left
MCP_cb_L	Middle Cerebellar Peduncle Cerebellar Part left
MCP_cb_R	Middle Cerebellar Peduncle Cerebellar Part right
MCP_L	Middle cerebellar peduncle left
MCP_R	Middle cerebellar peduncle right
Medulla_L	MEDULLA left
Medulla_R	MEDULLA right
MFG_DPFC_L	Middle frontal gyrus (dorsal prefrontal cortex) left
MFG_DPFC_R	Middle frontal gyrus (dorsal prefrontal cortex) right
MFG_L	MIDDLE FRONTAL GYRUS (posterior segment) left
MFG_R	MIDDLE FRONTAL GYRUS (posterior segment) right
MFOG_L	MIDDLE FRONTO-ORBITAL GYRUS left
MFOG_R	MIDDLE FRONTO-ORBITAL GYRUS right
MFOWM_L	MIDDLE FRONTO-ORBITAL WM left
MFOWM_R	MIDDLE FRONTO-ORBITAL WM right
MFWM_DPFC_L	Middle frontal WM (dorsal prefrontal cortex) left
MFWM_DPFC_R	Middle frontal WM (dorsal prefrontal cortex) right
MFWM_L	MIDDLE FRONTAL WM (posterior segment) left

MFWM_R	MIDDLE FRONTAL WM (posterior segment) right
Midbrain_L	MIDBRAIN left
Midbrain_R	MIDBRAIN right
ML_L	Medial lemniscus left
ML_R	Medial lemniscus right
MOG_L	MIDDLE OCCIPITAL GYRUS left
MOG_R	MIDDLE OCCIPITAL GYRUS right
MOWM_L	MIDDLE OCCIPITAL WM left
MOWM_R	MIDDLE OCCIPITAL WM right
MTG_L	MIDDLE TEMPORAL GYRUS left
MTG_L_pole	Pole of MIDDLE TEMPORAL GYRUS left
MTG_R	MIDDLE TEMPORAL GYRUS right
MTG_R_pole	Pole of MIDDLE TEMPORAL GYRUS right
MTWM_L	MIDDLE TEMPORAL WM left
MTWM_L_pole	Pole of MIDDLE TEMPORAL WM left
MTWM_R	MIDDLE TEMPORAL WM right
MTWM_R_pole	Pole of MIDDLE TEMPORAL WM right
Mynert_L	Nucleus innominata of Mynert left
Mynert_R	Nucleus innominata of Mynert right
NucAccumbens_L	Nucleus accumbens left
NucAccumbens_R	Nucleus accumbens right
OcciptSul_L	Occipital Lobe Sulci Left
OcciptSul_R	Occipital Lobe Sulci Right
OlfactoryRadiation_L	olfactory radiation left
OlfactoryRadiation_R	olfactory radiation right
OpticTract_L	Optic tract left
OpticTract_R	Optic tract right
ParietSul_L	Parietal Lobe Sulci Left
ParietSul_R	Parietal Lobe Sulci Right
PCC_L	posterior cingulate gyrus left
PCC_R	posterior cingulate gyrus right
PCCWM_L	posterior cingulate WM left
PCCWM_R	posterior cingulate WM right
PCR_L	Posterior corona radiata left
PCR_R	Posterior corona radiata right
PCT_L	Pontine crossing tract (a part of MCP) left
PCT_R	Pontine crossing tract (a part of MCP) right
PHG_L	PARAHIPPOCAMPAL GYRUS left
PHG_R	PARAHIPPOCAMPAL GYRUS right
PLIC_L	Posterior limb of internal capsule left
PLIC_R	Posterior limb of internal capsule right
PoCG_L	POSTCENTRAL GYRUS left
PoCG_R	POSTCENTRAL GYRUS right
PoCWM_L	POSTCENTRAL WM left
PoCWM_R	POSTCENTRAL WM right
Pons_L	PONS left
Pons_R	PONS right
PrCG_L	PRECENTRAL GYRUS left

PrCG_R	PRECENTRAL GYRUS right
PrCu_L	PRE-CUNEUS left
PrCu_R	PRE-CUNEUS right
PrCuWM_L	PRE-CUNEUS WM left
PrCuWM_R	PRE-CUNEUS WM right
PrCWM_L	PRECENTRAL WM left
PrCWM_R	PRECENTRAL WM right
PTR_L	Posterior thalamic radiation (include optic radiation) left
PTR_R	Posterior thalamic radiation (include optic radiation) right
Put_L	PUTAMEN left
Put_R	PUTAMEN right
PVWf_L	Periventricular White Matter frontal left
PVWf_R	Periventricular White Matter frontal right
PVWp_L	Periventricular White Matter posterior left
PVWp_R	Periventricular White Matter posterior right
PVWpl_L	Periventricular White Matter posterior lateral left
PVWpl_R	Periventricular White Matter posterior lateral right
RedNc_L	Red Nucleus left
RedNc_R	Red Nucleus right
RG_L	GYRUS RECTUS left
RG_R	GYRUS RECTUS right
RGWM_L	WM RECTUS left
RGWM_R	WM RECTUS right
RLIC_L	Retrolenticular part of internal capsule left
RLIC_R	Retrolenticular part of internal capsule right
rostral_ACC_L	rostral anterior cingulate gyrus left
rostral_ACC_R	rostral anterior cingulate gyrus right
rostralWM_ACC_L	rostral anterior cingulate WM left
rostralWM_ACC_R	rostral anterior cingulate WM right
SCC_L	Splenium of corpus callosum left
SCC_R	Splenium of corpus callosum right
SCP_L	Superior cerebellar peduncle left
SCP_R	Superior cerebellar peduncle right
SCR_L	Superior corona radiata left
SCR_R	Superior corona radiata right
SFG_L	SUPERIOR FRONTAL GYRUS (posterior segment) left
SFG_PFC_L	Superior frontal gyrus (prefrontal cortex) left
SFG_PFC_R	Superior frontal gyrus (prefrontal cortex) right
SFG_pole_L	superior frontal gyrus (frontal pole) left
SFG_pole_R	superior frontal gyrus (frontal pole) right
SFG_R	SUPERIOR FRONTAL GYRUS (posterior segment) right
SFO_L	Superior fronto-occipital fasciculus (could be a part of anterior internal capsule) left
SFO_R	Superior fronto-occipital fasciculus (could be a part of anterior internal capsule) right
SFWM_L	SUPERIOR FRONTAL WM (posterior segment) left
SFWM_PFC_L	Superior frontal WM (prefrontal cortex) left
SFWM_PFC_R	Superior frontal WM (prefrontal cortex) right
SFWM_pole_L	superior frontal WM (frontal pole) left
SFWM_pole_R	superior frontal WM (frontal pole) right

SFWM_R	SUPERIOR FRONTAL WM (posterior segment) right
SKULL1	
SKULL2	
SKULL3	
SLF_L	Superior longitudinal fasciculus left
SLF_R	Superior longitudinal fasciculus right
SMG_L	SUPRAMARGINAL GYRUS left
SMG_R	SUPRAMARGINAL GYRUS right
SMWM_L	SUPRAMARGINAL WM left
SMWM_R	SUPRAMARGINAL WM right
Snigra_L	Substancia Nigra left
Snigra_R	Substancia Nigra right
SOG_L	SUPERIOR OCCIPITAL GYRUS left
SOG_R	SUPERIOR OCCIPITAL GYRUS right
SOWM_L	SUPERIOR OCCIPITAL WM left
SOWM_R	SUPERIOR OCCIPITAL WM right
SPG_L	SUPERIOR PARIETAL GYRUS left
SPG_R	SUPERIOR PARIETAL GYRUS right
SPWM_L	SUPERIOR PARIETAL WM left
SPWM_R	SUPERIOR PARIETAL WM right
SS_L	Sagittal stratum (include inferior longitidinal fasciculus and inferior fronto-occipital fasciculus) left
SS_R	Sagittal stratum (include inferior longitidinal fasciculus and inferior fronto-occipital fasciculus) right
STG_L	SUPERIOR TEMPORAL GYRUS left
STG_L_pole	Pole of SUPERIOR TEMPORAL GYRUS left
STG_R	SUPERIOR TEMPORAL GYRUS right
STG_R_pole	Pole of SUPERIOR TEMPORAL GYRUS right
STWM_L	SUPERIOR TEMPORAL WM left
STWM_L_pole	Pole of SUPERIOR TEMPORAL WM left
STWM_R	SUPERIOR TEMPORAL WM right
STWM_R_pole	Pole of SUPERIOR TEMPORAL WM right
subcallosal_ACC_L	subcallosal anterior cingulate gyrus left
subcallosal_ACC_R	subcallosal anterior cingulate gyrus right
subcallosalWM_ACC_L	subcallosal anterior cingulate WM left
subcallosalWM_ACC_R	subcallosal anterior cingulate WM right
subgenual_ACC_L	subgenual anterior cingulate gyrus left
subgenual_ACC_R	subgenual anterior cingulate gyrus right
subgenualWM_ACC_L	subgenual anterior cingulate WM left
subgenualWM_ACC_R	subgenual anterior cingulate WM right
SylFrontSul_L	Sylvian Fissure Frontal Lobe Part Left
SylFrontSul_R	Sylvian Fissure Frontal Lobe Part Right
SylParieSul_L	Sylvian Fissure Parietal Lobe Part Left
SylParieSul_R	Sylvian Fissure Parietal Lobe Part Right
SylTempSul_L	Sylvian Fissure Temporal Lobe Part Left
SylTempSul_R	Sylvian Fissure Temporal Lobe Part Right
TAP_L	Tapatum left
TAP_R	Tapatum right

TempSul_L	Temporal Lobe Sulci Left
TempSul_R	Temporal Lobe Sulci Right
Thalamus_L	THALAMUS left
Thalamus_R	THALAMUS right
UNC_L	Uncinate fasciculus left
UNC_R	Uncinate fasciculus right